1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables

Highly relevant.

- Students must extract key numerical information from the text: "total climbers ≈ 200,000" and "open period: 1 July to 27 August."
- Significant variables include: total number of climbers (total quantity), number of open days (time span), and average daily climbers (target quantity).
- This is the foundational step in modeling: translating real-world language into mathematical elements.

2. Competency #2: Identify the key variables in a model

Highly relevant (closely linked to #6).

- The model clearly involves three variables: total climbers $T = 200{,}000$, number of days D , and average daily climbers $A = T/D$.
- Even though the model is simple, it still requires variable identification.

3. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

Moderately relevant.

- Students must simplify the complex real-world scenario of a "climbing season" into a countable number of days.
- They also assume that varying daily numbers can be represented by a single average—a standard modeling simplification.

4. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

Moderately relevant.

- Implicit assumptions include: all climbers ascend during the official period; daily variation can be ignored via averaging; and the date range is calculated accurately.
- While the question doesn't explicitly ask students to state these, recognizing them is essential for valid modeling.

5. Competency #1: Select a mathematical description or representation that describes a problem

Slightly relevant.

- Students choose the formula "average = total ÷ number of days" as the appropriate mathematical description.
- Though simple, this still involves selecting a suitable mathematical structure.

6. Competency #9: Translate a problem into a standard mathematical representation or algorithm

Slightly relevant.

- The word problem is converted into the arithmetic expression: $\text{Average} = 200{,}000 \div (\text{number of days})$.
- This is a straightforward application of a standard division algorithm.

7. Competency #3: Select a representation appropriate to the problem context

Low relevance.

- There is essentially only one natural representation here (arithmetic mean), so no meaningful choice among multiple representations (e.g., graphs, equations, ratios) is required.

1. 第6项：Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

高度相关。

- 学生需要从文本中识别出关键数学信息：“总人数 ≈ 200,000” 和 “开放天数（7月1日到8月27日）”。
- 显著变量是：总攀登人数（总量）、开放天数（时间跨度）、每日平均人数（所求量）。
- 这是解题的第一步，也是建模的核心：从现实语言中提取数学要素。

2. 第2项：Identify the key variables in a model

高度相关，与第6项紧密关联。

- 明确模型中的三个变量：总人数 $T = 200{,}000$，天数 D，日均人数 $A = T/D$。
- 虽然模型简单，但确实涉及变量识别。

3. 第7项：Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

中等相关。

- 学生需将“登山季节”这一复杂现实简化为一个时间段，并进一步简化为“总天数”。
- 同时，将“每天人数不同”的现实简化为“平均每天人数”，忽略波动，这是典型的建模简化假设。

4. 第12项：Identify the constraints, assumptions, simplifications in a mathematical model

中等相关。

- 隐含假设包括：所有登山者都在开放期内登山；每日人数可被平均化；日期计算准确（不含误差）。
- 虽然题目未明确要求学生陈述这些假设，但理解这些是正确建模的前提。

5. 第1项：Select a mathematical description or a representation that describes a problem

中低相关。

- 学生选择“平均数 = 总量 ÷ 天数”这一数学表达式来描述问题。
- 虽然模型简单，但仍属于选择适当数学结构的过程。

6. 第9项：Translate a problem into a standard mathematical representation or algorithm

中低相关。

- 将文字问题转化为算式：$\text{Average} = 200{,}000 \div \text{number of days}$。
- 这是一个标准的除法算法应用。

7. 第3项：Select a representation appropriate to the problem context

低相关。

- 此题几乎只有一种自然表示方式（算术平均），无需在多种表示（如图表、方程、比例等）间选择。